```
#include <stdio.h>
int main() {
    int n, quantum, time = 0, completed = 0;
    int queue[100], front = 0, rear = 0;
    int visited[100] = {0};
    printf("1WA24CS218\n");
    printf("Enter number of processes: ");
    scanf("%d", &n);
    int at[n], bt[n], rt[n], ct[n], tat[n], wt[n];
    for (int i = 0; i < n; i++) {
        printf("Enter AT and BT for P%d: ", i + 1);
        scanf("%d %d", &at[i], &bt[i]);
        rt[i] = bt[i];
    }
    printf("Enter Time Quantum: ");
    scanf("%d", &quantum);
    for(int i=0; i<n; i++) {
        if(at[i] <= time) {
            queue[rear++] = i;
            visited[i] = 1;
        }
    }
    while (front < rear) {
       int i = queue[front++];
        int execute = (rt[i] > quantum) ? quantum : rt[i];
        rt[i] -= execute;
        time += execute;
        for (int j = 0; j < n; j++) {
            if (at[j] <= time && visited[j] == 0) {
                queue[rear++] = j;
```

```
                visited[j] = 1;
            }
        }
        if (rt[i] > 0) {
            queue[rear++] = i;
        } else {
            ct[i] = time;
            completed++;
        }
        if (front == rear && completed < n) {
            for (int j = 0; j < n; j++) {
                if (visited[j] == 0) {
                    time = at[j];
                    queue[rear++] = j;
                    visited[j] = 1;
                    break;
                }
            }
        }
    }
    printf("\nID\tAT\tBT\tCT\tTAT\tWT\n");
    for (int i = 0; i < n; i++) {
        printf("%d\t%d\t%d\t%d\t%d\t%d\n", i+1, at[i], bt[i], ct[i], ct[i]-at[i], (ct[i]-at[i])-bt[i]);
    }
    return 0;
}
```

```
1WA24CS218
Enter number of processes: 5
Enter AT and BT for P1: 0 5
Enter AT and BT for P2: 1 3
Enter AT and BT for P3: 2 1
Enter AT and BT for P4: 3 2
Enter AT and BT for P5: 4 3
Enter Time Quantum: 2

ID      AT      BT      CT      TAT     WT
1       0       5       13      13      8
2       1       3       12      11      8
3       2       1       5       3       2
4       3       2       9       6       4
5       4       3       14      10      7

Process returned 0 (0x0)   execution time : 26.236 s
Press any key to continue.
```